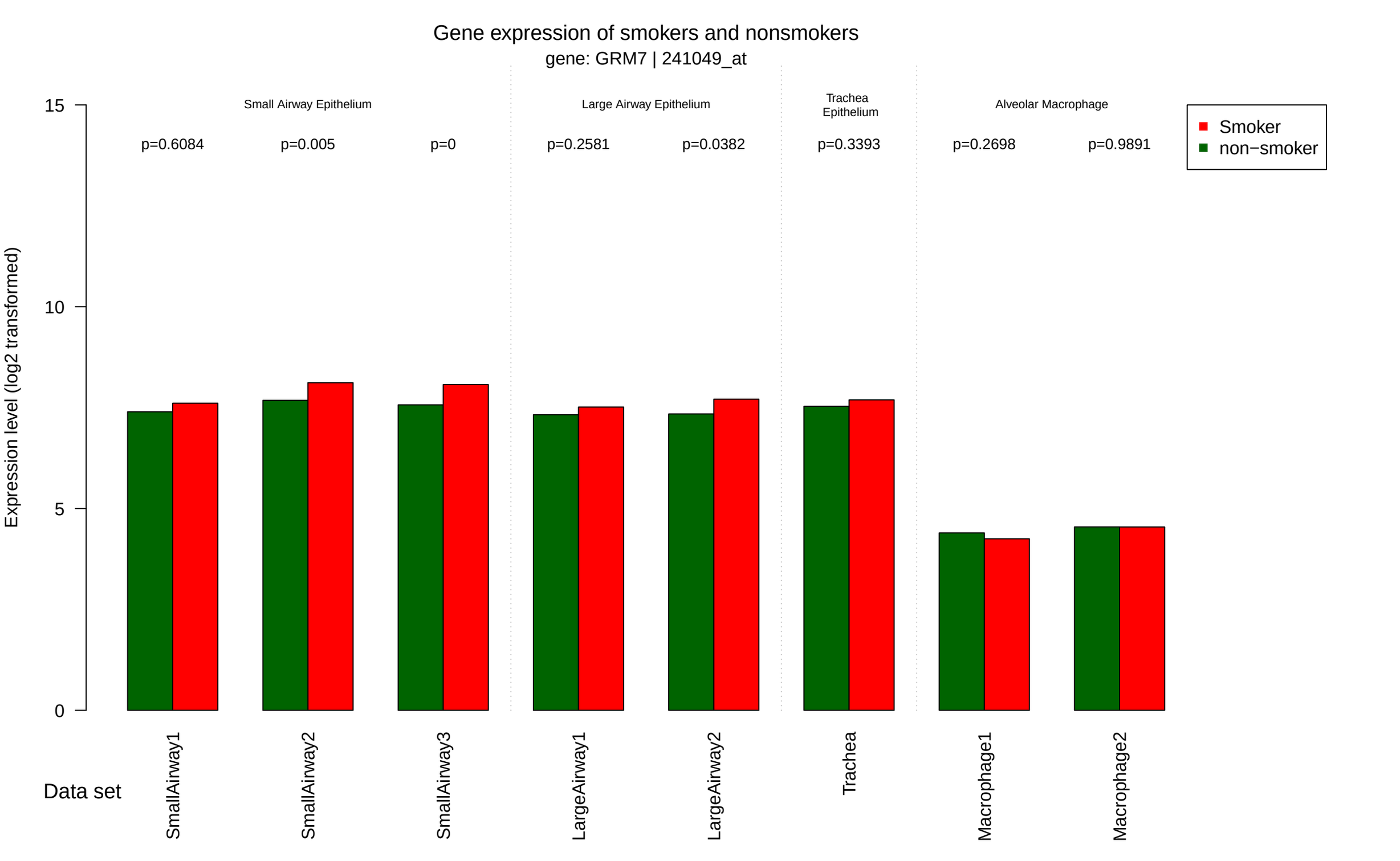
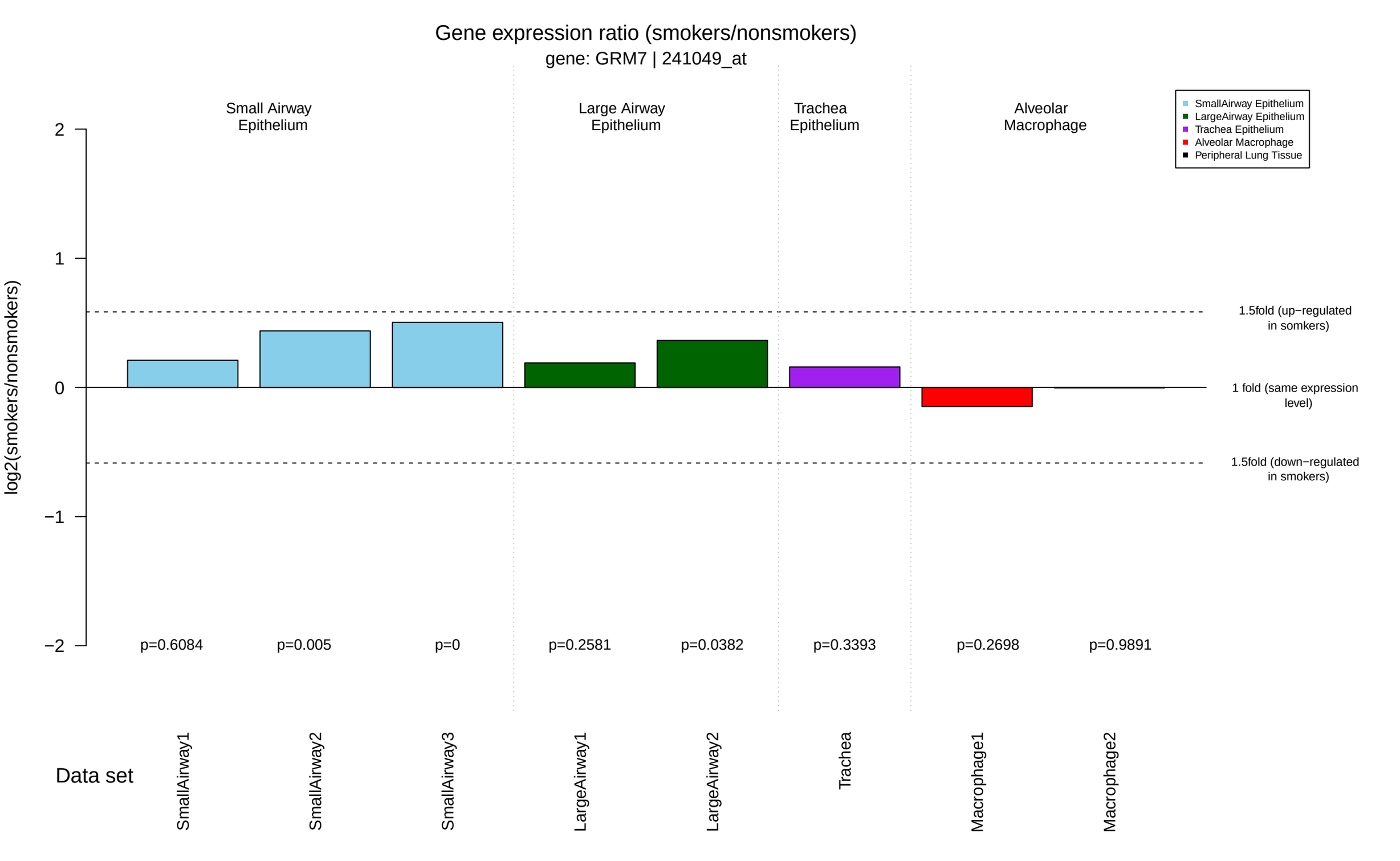
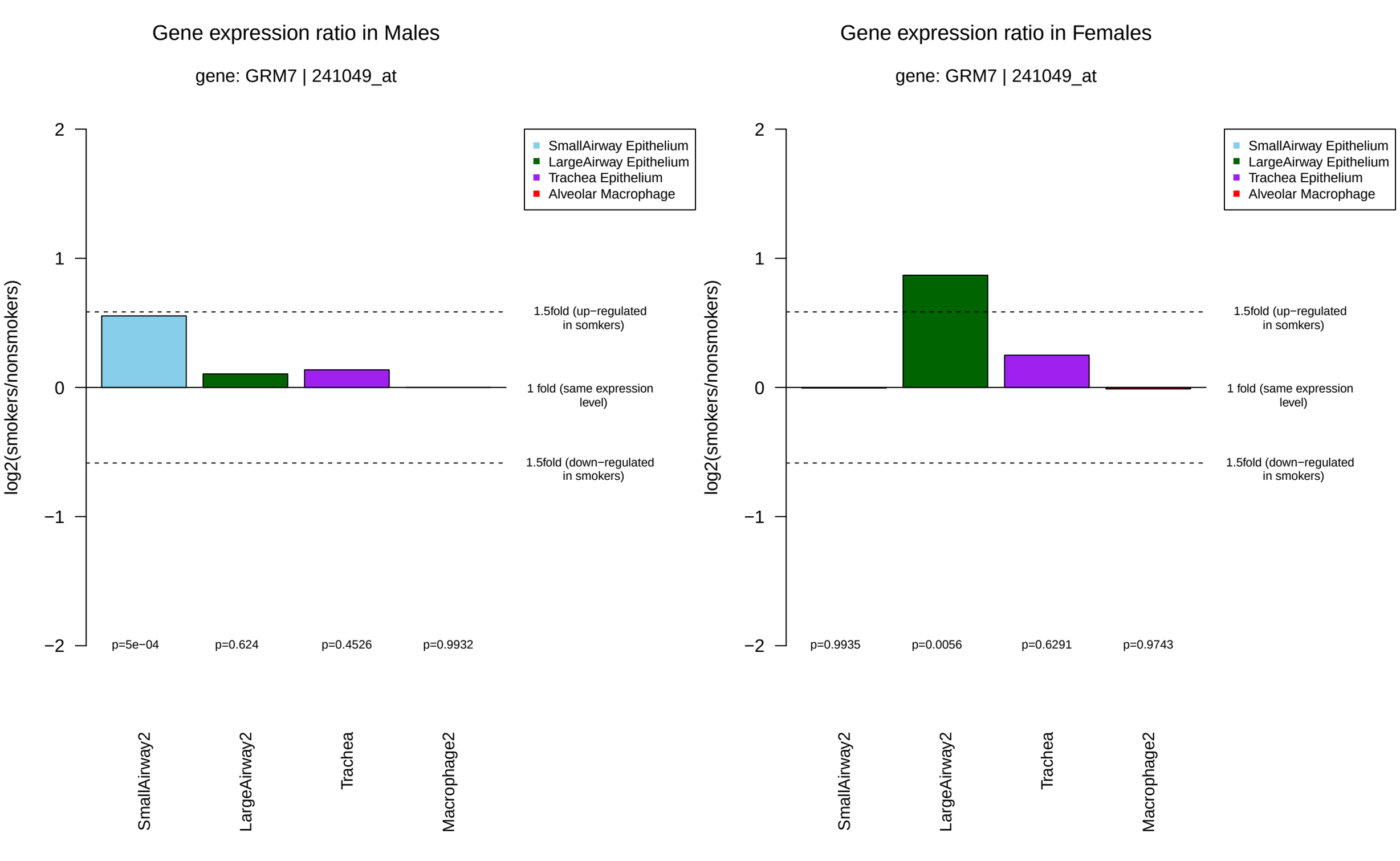
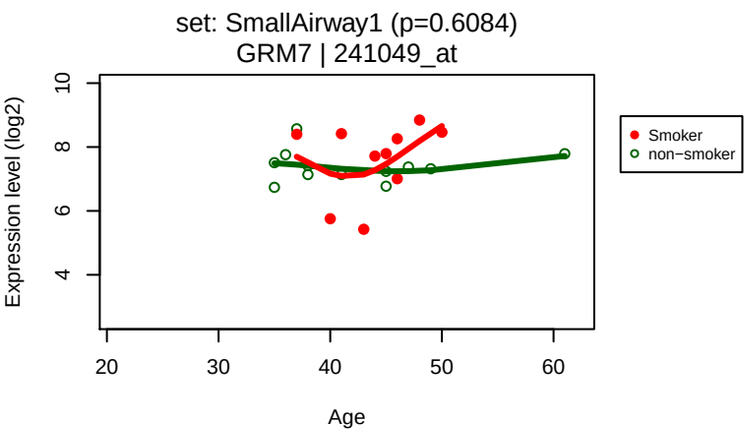
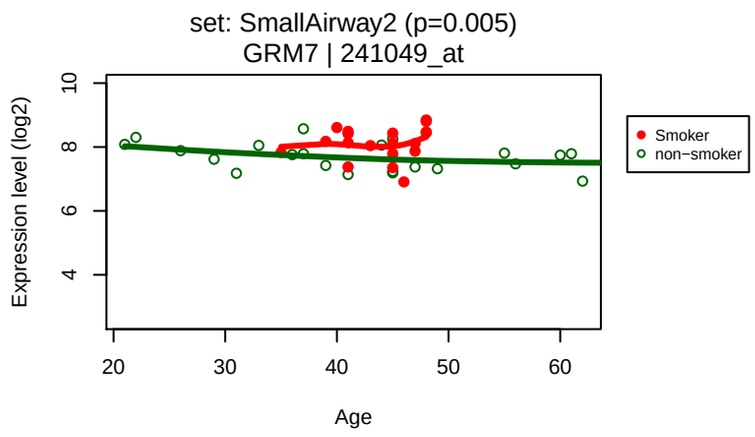
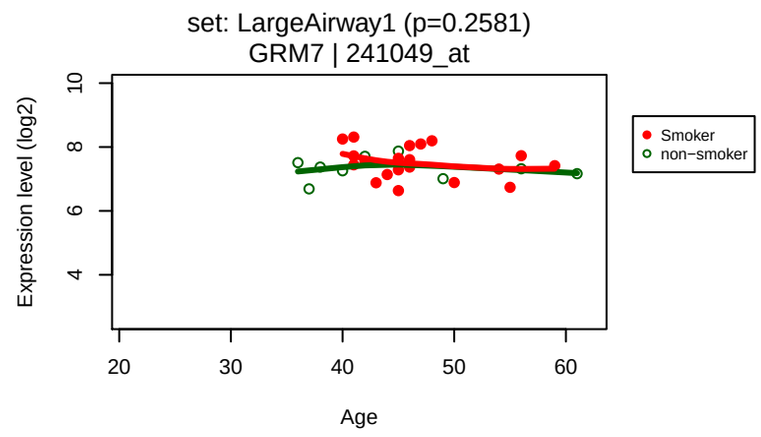

Gene expression of smokers and nonsmokers
gene: GRM7 | 241049_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6084
p=0.005
p=0
p=0.2581
p=0.0382
p=0.3393
p=0.2698
p=0.9891
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

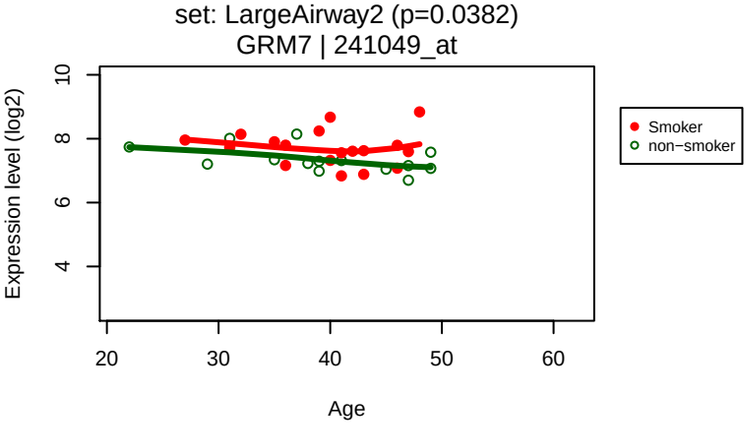

Gene expression ratio (smokers/nonsmokers)
gene: GRM7 | 241049_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6084
p=0.005
p=0
p=0.2581
p=0.0382
p=0.3393
p=0.2698
p=0.9891
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

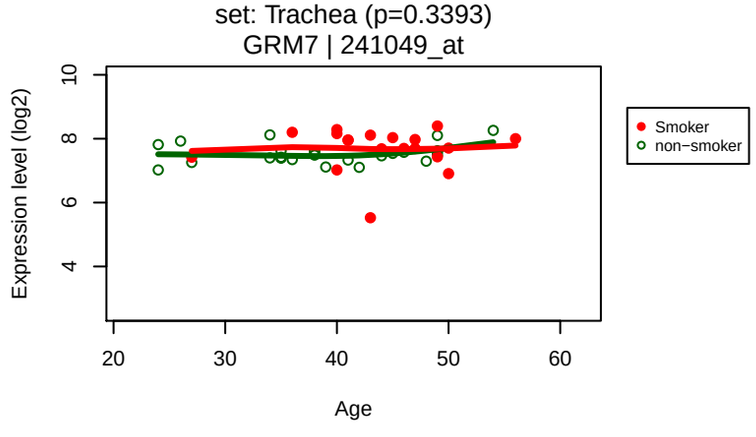

Gene expression ratio in Males
gene: GRM7 | 241049_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=5e−04
p=0.624
p=0.4526
p=0.9932
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRM7 | 241049_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9935
p=0.0056
p=0.6291
p=0.9743
SmallAirway2
LargeAirway2
Trachea
Macrophage2

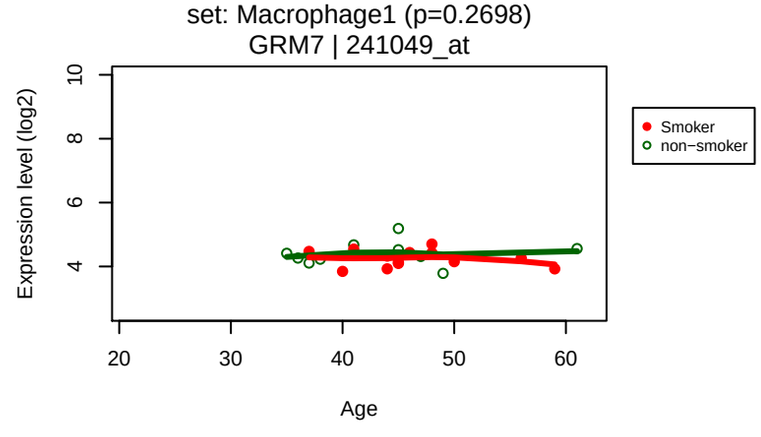

set: SmallAirway1 (p=0.6084)
GRM7 | 241049_at
Expression level (log2)
Age
Smoker
non−smoker

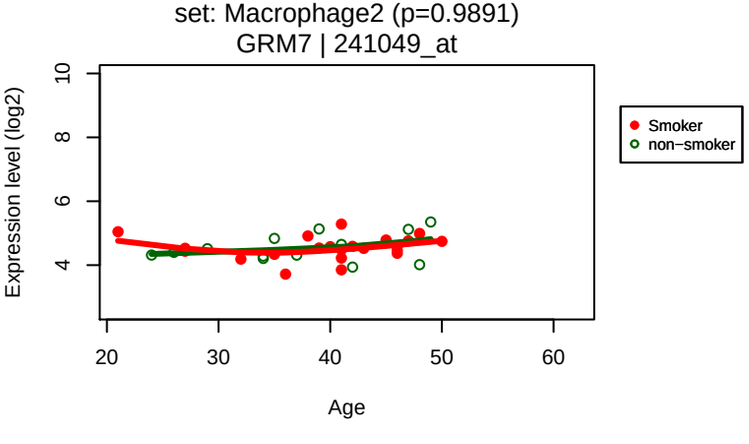

set: SmallAirway2 (p=0.005)
GRM7 | 241049_at
Expression level (log2)
Age
Smoker
non−smoker

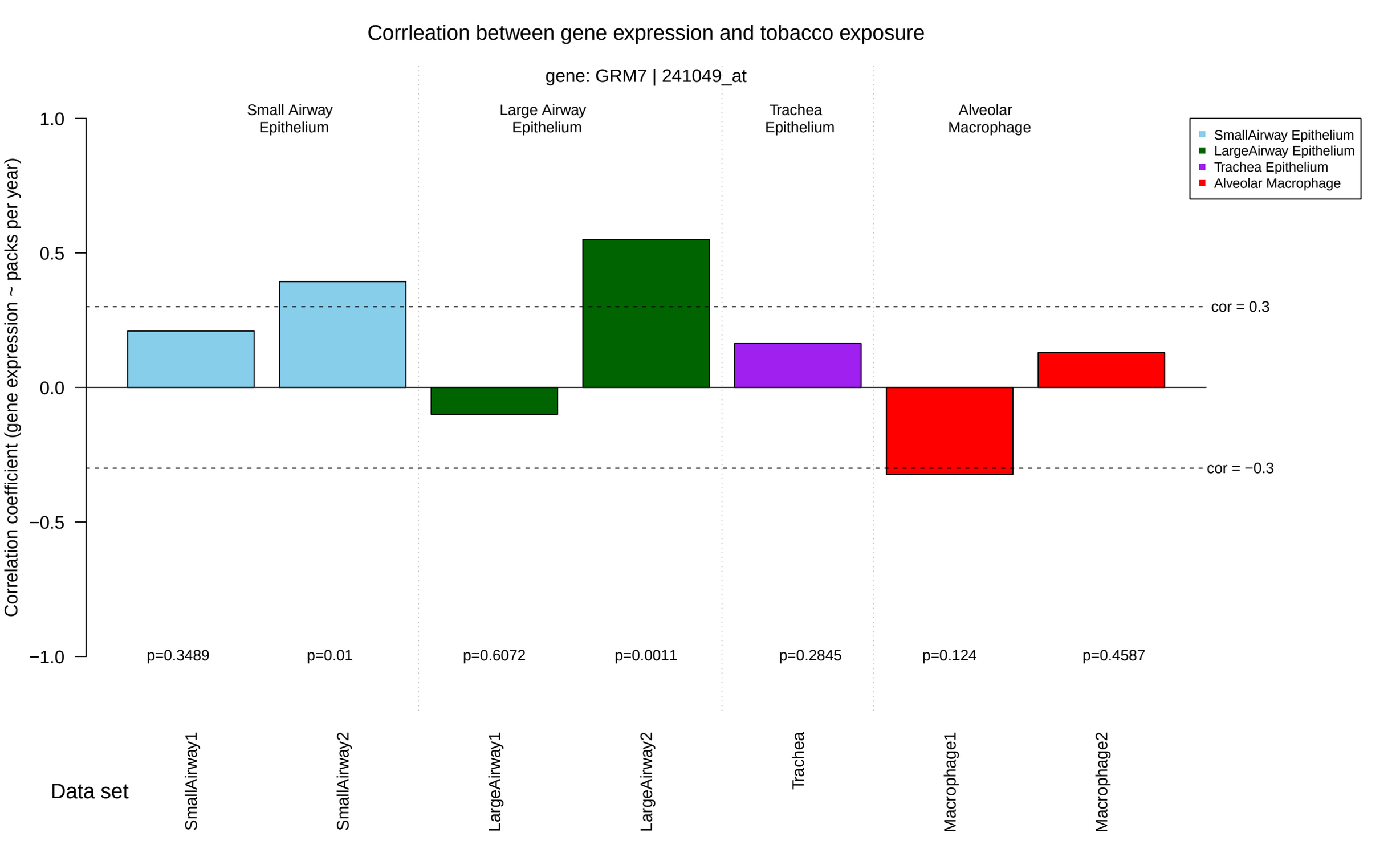

set: LargeAirway1 (p=0.2581)
GRM7 | 241049_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0382)
GRM7 | 241049_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3393)
GRM7 | 241049_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2698)
GRM7 | 241049_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9891)
GRM7 | 241049_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRM7 | 241049_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3489
p=0.01
p=0.6072
p=0.0011
p=0.2845
p=0.124
p=0.4587
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2